A Paired-Delta QRNG Architecture for Controlling Shared Artifacts in Observer–Randomness Experiments

By Andrea Rester Campbell

Independent Researcher

Port Angeles, WA USA

Abstract

Experiments testing whether human intention influences quantum random number generators (QRNGs) face two persistent problems. The first is one of credibility: across studies, small reported effects are difficult to separate from publication bias, selective reporting, and analytic flexibility. The second is one of within-study artifact ambiguity: when expected deviations are small relative to system noise, a single-channel design cannot distinguish a proposed observer-linked effect from hardware drift, timing artifacts, or environmental fluctuation, and increasing statistical power amplifies systematic confounds alongside any genuine signal. This paper addresses the second problem with a structural design, and the first with pre-registration and data-freezing.

We describe a Paired-Delta Protocol (PDP) that addresses this problem by building hardware control into each QRNG call rather than relying on external baselines. Each block generates a single 301-bit quantum call, divided by a quantum-random assignment bit into a Subject stream (observer-aligned) and a Paired Control Stream (PCS). Because both streams originate from the same physical draw, any shared hardware or environmental fluctuation is expected to affect them similarly and therefore cancel under subtraction (Subject − PCS), providing common-mode rejection at the level of each trial for artifacts that act similarly on both streams.

We report validation results from pilot data (121 participants; 11,607 blocks across human, AI, and baseline conditions). First, the paired null distribution is empirically validated: the two halves of each call are statistically independent under hardware-only conditions. Second, artifact-injection tests confirm that the architecture cancels additive common-mode artifacts by 97.8–99.7% (metric-level offsets) and 71–94% (bit-level bias), while stream-specific effects survive at the level predicted by variance scaling. Third, the architecture provides a dual-analysis property: because the Subject and PCS streams derive from the same call at no additional cost, every call yields both a paired-delta analysis (artifact-immune) and a direct single-stream analysis (full statistical power), which are complementary rather than redundant.

As an illustrative application of the architecture, we apply a within-block temporal-ordering metric (the single-scale Hurst exponent) to the pilot dataset. This exploratory analysis surfaces candidate ordering structure in a small high-engagement subgroup; the finding is preliminary and is presented to demonstrate the kind of structure the architecture can surface, not as a confirmatory result. A pre-registered replication has been designed to test whether the pattern generalizes.

The primary contribution is the methodological architecture itself: a design in which any proposed deviation must survive within-call hardware control, cross-condition comparison, and temporal diagnostics before interpretation.

1. Introduction

1.1 The Credibility Problem in Observer–QRNG Research

Decades of experiments testing whether human intention influences quantum or electronic random number generators have produced a large but contested body of data (e.g., Jahn et al., 1997). The central scientific disputes in this literature are primarily about the credibility of the aggregate evidence. The most influential skeptical analysis, Bösch, Steinkamp, and Boller's (2006) meta-analysis of 380 studies in Psychological Bulletin, found a statistically significant but extremely small overall effect that was strongly and inversely related to sample size and highly heterogeneous. They argued it is a pattern consistent with publication bias and selective reporting rather than a genuine effect. The ensuing exchange (Radin, Nelson, Dobyns, & Houtkooper, 2006; Wilson & Shadish, 2006) disputed the interpretation but not the underlying concern: that small effects in this paradigm are difficult to separate from the combined influence of analytic flexibility, selective reporting, and publication bias. More recent pre-registered work has reported Bayesian evidence against micro-psychokinetic effects on QRNG output (Maier, Dechamps, & Pflitsch, 2018), reinforcing that the field's central unresolved question is one of credibility and reproducibility, not detection per se.

This points to two distinct problems that any single study in this paradigm must confront.

1.1.1 Problem 1: Cross-study credibility

The dominant critiques concern the research process: undisclosed analytic choices, outcome-dependent stopping, and selective publication, which together can manufacture small apparent effects across a literature. The accepted remedies are procedural — pre-registration of hypotheses and analyses, locked datasets, and transparent reporting — applied before outcomes are known.

1.1.2 Problem 2: Within-study artifact ambiguity

Separately, within any individual study, when expected deviations are small relative to system noise, a single-channel design cannot cleanly determine whether an observed deviation reflects an observer-linked effect or ordinary variation in hardware behavior (device drift, temperature, timing, or environmental fluctuation). Increasing sample size does not resolve this; it raises sensitivity to genuine signals and systematic confounds alike. What is required is a design that separates shared hardware fluctuations from stream-specific deviations at the point of data generation rather than through post-hoc adjustment.

This paper addresses both problems, though with responses of different scope and completeness. For Problem 1 (credibility) the pilot's safeguards include: metrics were pre-specified in the experiment code before data collection, and the dataset was frozen with cryptographic fingerprints prior to analysis. All findings are treated as exploratory. The complete remedy for this problem is prospective pre-registration, which this pilot motivates but does not itself provide (Section 4.4).

For Problem 2 (within-study artifact ambiguity) we introduce a paired-delta architecture that builds a simultaneous hardware control into every observation (Section 1.2). This solution is structural and complete: implemented at the data-generation level, requiring no post-hoc adjustment, and designed before data collection began.

Beyond artifact control, the architecture also enables analysis of within-block ordering structure, a property of binary streams that is analytically distinct from hit rate and invisible to single-channel designs. This capability is illustrated in Section 1.3.

1.2 The Paired-Delta Solution

The Paired-Delta Protocol addresses the artifact problem by building a simultaneous hardware control into every observation. The key innovation is structural: rather than comparing a participant's stream to an external baseline collected at a different time, every experimental block generates its own internal reference from the same quantum call.

Each block proceeds as follows:

1. A single QRNG call generates 301 quantum-random bits.
2. Bit 0 serves as a quantum-random assignment bit.
3. Bits 1-150 become halfA; bits 151-300 become halfB.
4. If the assignment bit = 1: Subject = halfA, PCS = halfB. If the assignment bit = 0: Subject = halfB, PCS = halfA.
5. On each block, a randomly assigned target was displayed. Participants initiated a single 301-bit QRNG call, during which the target pulsed at 5 Hz. Feedback was based solely on the Subject stream; the paired control stream (PCS) was never displayed and served as the hardware reference.
6. After every fifth block, a separate 1000-bit QRNG audit call is made without any task context (no target display, no intention, no feedback). These audit samples provide independent hardware validation throughout each session.
7. All primary within-call statistics are computed as paired differences:

$$\Delta_M = M_{Subject} - M_{PCS}$$

where M may represent hit rate, Hurst exponent, or any other stream-level metric computed on the 150-bit sequences. Because both streams derive from the same physical draw, any hardware drift, temperature fluctuation, or environmental noise affects both halves identically and cancels under subtraction.

Importantly, this paired subtraction is structural rather than statistical: the two streams share the same physical origin and differ only by label assignment. As shown in Figure 1, this achieves common-mode rejection at the level of each trial ; the paired design is not hostile to subject effects, but hostile to artifacts.

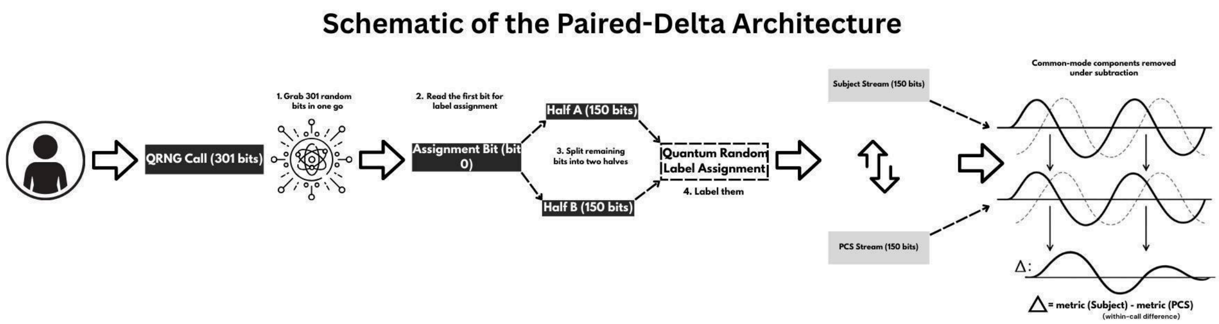

Figure 1. Paired-Delta QRNG architecture. A single QRNG call generates 301 bits. Bit 0 serves as a quantum-random assignment bit. The remaining 300 bits are divided into two 150-bit halves. Based on the assignment bit, one half is labeled the Subject stream and the other the Paired Control Stream (PCS). All paired statistics are computed as within-call differences ($\Delta_M = M_{Subject} - M_{PCS}$), thereby cancelling components shared by both halves of the same physical draw.

The design has four properties relevant to artifact control.

Simultaneity. Both streams come from the same QRNG call, the same chip state, the same microsecond. Any hardware drift, temperature effect, or environmental noise affects both streams identically.

Quantum label assignment. The assignment bit is itself a quantum random output. The hardware cannot "know" which half will be labeled Subject. There is no physical mechanism by which system-level artifacts could systematically favor one label over the other.

Common-mode rejection. The paired difference $(Subject - PCS)$ cancels any signal shared by both streams (analogous to differential signaling in electronics, where common-mode noise is subtracted out to isolate the true signal), while leaving stream-specific deviations intact. If an effect acts specifically on the stream aligned with the observer, it does not cancel under pairing. The design removes shared artifacts while allowing stream-specific deviations to survive at a level predicted by variance scaling (Section 3.3.3). The paired design is therefore not hostile to subject effects (any such effect remains detectable, at the cost of a $\sqrt{2}$ variance penalty), but it is hostile to artifacts, which are cancelled far more completely (Section 3.3).

Dual-analysis at no additional cost. Because Subject and PCS derive from the same call, every call yields two complementary analyses for the price of one: the paired-delta (artifact-immune, at the cost of a $\sqrt{2}$ variance increase) and the direct single-stream analysis (full power, no variance cost). These are not independent tests (they share the same underlying data), but they are analytically distinct and answer different questions. When hardware is clean, the direct analysis carries the sensitivity; when artifacts are present, the delta carries the validity. Agreement between them strengthens a finding; divergence localizes whether observed structure is stream-specific (surviving in the delta) or shared across the call (visible directly but cancelled in the delta). No single-channel design can make this distinction, because it has nothing to compare against. The implications of this property for analysis design are developed in Section 4.3.

A central design question was how to distinguish between different kinds of possible effects. If something were influencing the QRNG output during human participation, would it affect the entire call uniformly, or would it depend on which half of the call was labeled as the Subject stream? We did not assume one or the other. The architecture was built so that these alternatives could be separated empirically.

Pairing eliminates whole-call shifts within a single QRNG draw: any uniform influence (whether arising from hardware drift or, hypothetically, from an observer effect acting on the entire call rather than a specific stream) cancels under subtraction.

This is a structural limitation of the architecture: a whole-call effect and a whole-call artifact are indistinguishable from the paired delta alone. However, broader session-level effects would be detectable through analysis of the control (PCS) stream itself across conditions (Section 3.1): if intent acted on the whole call uniformly, PCS during Human sessions would be expected to differ from PCS during Baseline sessions, since both Subject and PCS would be affected equally during Human sessions but not during Baseline.

No such difference was observed; PCS hit rate and Hurst exponent showed no significant differences across Human, AI, and Baseline conditions ($p > 0.50$). Independent audit calls (Section 3.4) and post-hoc reassignment of stream labels provide additional diagnostics. Together, these allow differentiation between shared hardware effects, session-level drift, and stream-specific deviations, though they cannot, by construction, rule out a whole-call effect that is itself indistinguishable from hardware-level uniformity.

1.3 Beyond Artifact Control: Analyzing Within-Block Structure

The architecture's value is not limited to artifact rejection. Because it preserves the full bit sequence of each stream, it supports analyses of within-block ordering structure (the persistence or alternation of bits within a block) that single-channel mean-based designs cannot evaluate cleanly. Ordering structure is analytically independent of bit composition: a sequence may match expected frequencies exactly while exhibiting non-random ordering, and a metric sensitive to ordering therefore provides a diagnostic dimension orthogonal to the hit rate.

To illustrate this capability, we pre-specified a temporal-ordering metric, the single-scale Hurst exponent, and applied it to the pilot dataset. It is introduced here only as one example of the distributional analyses the architecture supports; it is utilized as a screening signal whose flags are interrogated by convergent order-based diagnostics (within-block shuffling and label-swap permutation). The metric's definition, the rationale for the single-scale variant at $n = 150$, and its validation are given in Methods (Section 2.6.2); its application to the pilot data, and the exploratory pattern it surfaced, are reported in Section 3.6.

2. Methods

2.1 Participants

A total of 121 human participants contributed data. Recruitment occurred through two main channels plus a small amount of personal outreach.

Most participants (approximately 100) were recruited through a survey-exchange platform where users complete other researchers' studies in exchange for credits to post their own. This group participated primarily for platform incentives and had no special reason to care about the topic beyond completing the task.

A smaller group was recruited through informal posts to Facebook and to the Reddit community r/remoteviewing. Facebook recruitment produced only a handful of participants, while the Reddit community provided most of the non–survey-exchange participants. In the Reddit outreach, participants were explicitly encouraged to complete five sessions, because repeated participation was expected to be informative in a short-form protocol.

No exclusion criteria were applied prior to data collection.

Engagement was self-selected. Most participants completed a single session, a small number completed multiple sessions, and three participants completed five or more sessions. One of the three ≥5-session participants was the principal investigator, and contributed the most sessions (18) in that subgroup. For transparency, analyses that rely on the ≥5-session subgroup are treated as exploratory and are reported with leave-one-out checks to show whether results depend on any single individual.

Because recruitment source and engagement level are confounded in this pilot dataset, session-count effects cannot be interpreted as learning or “dose response.” Higher-session participants were disproportionately drawn from the Reddit recruitment, while single-session participants were disproportionately drawn from the survey-exchange platform. Separating repeated exposure from population self-selection requires a replication design that recruits both sources with balanced repeat-participation targets.

Intake data collected prior to each session included age and gender; ADHD and autism-spectrum diagnoses; a set of attention-related self-reports (daily screen time, frequency of attentional lapses, time in deep-focus activities, and frequency of flow states); and psi- and state-related items (meditation frequency, openness to psi possibility, and current mental clarity). Of these, meditation frequency and openness to psi possibility (the sheep–goat variable, contrasting psi believers with skeptics) were analyzed as exploratory moderators of hit rate and Hurst-delta (ordering-structure) metrics. The remaining items were collected for sample characterization and are reserved for analysis in the pre-registered replication, where the larger sample provides adequate power; they are available in the public dataset. Post-session ratings of focus, calm, and confidence, along with optional free-text feedback, were recorded but not analyzed.

2.2 Apparatus

Quantum random bits were generated using hardware quantum random number generators (QRNGs) accessed via public web APIs. Due to API rate limits and service availability constraints, three independent QRNG services were used sequentially during data collection: Outshift (Cisco Quantum Entropy Service), LFDR QRNG (Laboratory for Digitalisation Research; Germany), and the ANU Quantum Random Numbers Server (Australian National University). The majority of blocks were collected using the first two providers.

Each experimental block fetched 301 bits in a single API call. Both the Subject stream and the Paired Control Stream (PCS) for that block were derived from the same 301-bit call, ensuring that any device-specific characteristics affected both streams equally and cancelled under paired subtraction.

QRNG provider identity was not stored at the block level and therefore was not included as an analytic factor. However, provider transitions occurred due to service availability rather than experimental condition, and the paired-delta architecture mitigates device-level effects by performing within-call subtraction.

The experiment ran in a web browser. For human participants, a static target color was displayed. Once the participant initiated the QRNG call, the target pulsed at 5 Hz during bit generation.

2.3 Procedure

Each session consisted of 30 blocks. On each block, a target color was randomly assigned and displayed. When ready, the participant pressed a button to initiate a single 301-bit QRNG call. During bit generation, the target pulsed at 5 Hz while the participant maintained focus.

Bit 0 served as the assignment bit, randomly determining which 150-bit half of the call was labeled the Subject stream and which was labeled the Paired Control Stream (PCS). Participants never saw the raw bitstream. After completion of the call, feedback was displayed as the hit rate of the Subject stream (proportion of bits matching the intended direction). The PCS was scored identically but not displayed.

Every fifth block, a separate 1000-bit audit call was made without task context (no target, no intention, no feedback). These audit samples provided ongoing hardware validation independent of the experimental task.

2.4 Conditions

Three conditions were run:

2.4.1 Human

Participants performed the intention task as described above.

2.4.2 AI Agent

The experiment included an AI-driven condition implemented using GPT-4o-mini. This condition served two purposes. First, it provided a procedural control for task flow, timing, and interface dynamics under the paired-delta architecture. Second, it allowed an exploratory assessment of whether a non-biological agent, when given intent-framed prompts and feedback, would exhibit any measurable divergence from hardware baseline or any learning-like dynamics across blocks or sessions.

The AI condition consisted of two components: GPT-4o-mini, which generated text responses to prompts, and a Puppeteer automation layer that executed the corresponding browser actions. On each block, Puppeteer detected the “Ready” screen and provided GPT with a prompt mirroring what a human participant would see (e.g., “Focus intention on BLUE. Ready.”). After GPT returned a response, Puppeteer initiated the QRNG call. During the “Fetching quantum data…” phase, GPT received a second prompt aligned with the pulsing target display. After each block, the agent received a reduced form of the feedback available to human participants (its hit-rate percentage for that block) before proceeding.

Despite this alignment, the AI condition differs from human participation in important ways. Human participants are self-paced and may delay initiation, adjust their internal state, or pause between blocks. GPT, by contrast, operates within the lifecycle of an API call and cannot intentionally delay or modulate timing beyond the duration of its response generation. Although the agent’s output gates the initiation of each QRNG call, it does not exercise open-ended control over when initiation occurs. This implementation represents the closest analogue achievable within a stateless API-driven framework, while acknowledging that it is not equivalent to human self-pacing.

In the present study, the AI condition therefore functions both as a procedural control and as an exploratory probe of whether intent-framed prompting in a non-biological system produces any detectable structure under the paired-delta architecture.

2.4.3 Baseline

Fully automated execution with no agent present to act on the intent framing. Block initiation and inter-trial timing were scripted rather than self-paced.

2.5 Data Freeze

All data were frozen prior to analysis. SHA-256 fingerprints were computed for each file and stored in a manifest. All reported analyses use this frozen dataset.

After removing blocks with missing data or sessions containing fewer than 25 blocks, the final dataset comprised 11,607 blocks (human: 4,737; AI: 3,780; baseline: 3,090).

2.6 Pre-Specified Metrics

Two metrics were pre-specified in the experiment code:

2.6.1 Hit rate

The proportion of Subject bits matching the intended direction (hits / 150). The paired-delta hit-rate measure was computed within each QRNG call as:

$$\Delta_{rate} = rate_{Subject} - rate_{PCS}$$

This subtraction isolates within-call asymmetry between the observer-aligned stream and its simultaneous hardware control. Because both streams are derived from the same 301-bit QRNG draw, any whole-call shift in device behavior affects them equally and cancels under subtraction. The paired-delta formulation therefore tests for stream-specific deviation rather than deviation from an assumed ideal baseline of 0.5.

2.6.2 Single-scale Hurst exponent

The single-scale Hurst (H) exponent was pre-specified as an ordering-sensitive screening metric. It is not a second primary outcome alongside hit rate; rather, it serves as a flag for within-block temporal structure whose signals are interrogated by convergent diagnostics (within-block shuffling and label-swap permutation). Elevated ΔH values direct further inquiry; they do not constitute confirmatory evidence on their own. Unlike hit rate, which depends only on aggregate bit frequency, H captures persistence and clustering in the ordering of bits while remaining insensitive to overall composition. These are analytically distinct properties: the same 150 bits arranged in a different order generally produce a different H but the identical hit rate.

The Hurst exponent (H) is a classical measure of long-range dependence in time-series analysis, summarizing whether a process tends to persist (values cluster, $H > 0.5$) or anti-persist (values alternate, $H < 0.5$). Across hydrology, economics, and physics it has been used to quantify memory and scaling behavior in empirical time series.

The standard estimation approach computes the rescaled range (R/S) at multiple sub-window sizes and fits a regression through those points. This works well for long records, but for 150-bit blocks it is unreliable: (R/S) values on 10–25-bit sub-windows are quantized and noisy, and the regression slope becomes dominated by these unstable small scales (cf. Hamed, 2007). We therefore used a single-scale variant: one R/S calculation over the entire 150-bit block. To formalize the metric, raw bits were first mapped to a bipolar sequence $x \in \{1, -1\}$. The H was calculated as:

$$H = \frac{\log(R/S)}{\log(n)},\ n = 150$$

For each stream (Subject and PCS), the computation proceeded as follows: the cumulative deviation from the block mean was calculated across all 150 bits in sequence; R was taken as the range of that cumulative deviation trajectory (maximum minus minimum); S was the standard deviation of the 150-bit sequence;

and H was obtained from the formula above. Values were constrained to [0, 1], consistent with the theoretical limits of the Hurst exponent.

R is not the range of the raw bit values; it is the range of the cumulative deviation process. Each bit's deviation from the block mean is accumulated sequentially, producing a running trajectory across the block, and R is the difference between that trajectory's highest and lowest points. Because the trajectory depends on the order in which bits appear, the same 150 values arranged differently generally produce a different R. S, by contrast, is order-invariant. The single-scale Hurst estimate is therefore order-sensitive even though bit composition is unchanged: within-block shuffling alters H by disrupting the cumulative path while preserving marginal frequencies. This makes the shuffle analysis in Section 3.6.3 a diagnostic of ordering dependence rather than bit composition.

For a purely random sequence, H tends toward ~0.5 with a small positive finite-sample offset and is a known property of R/S on short records that is roughly constant for fixed n and therefore cancels in within-experiment comparisons. Values above 0.5 indicate longer runs than chance predicts; values below 0.5 indicate more frequent reversals. The estimator was pre-specified in the experiment's JavaScript source (implemented as the hurstApprox function) and computed on every block during data collection. Because the 150-bit blocks are short by time-series standards, we first validated that the single-scale variant responds to ordering structure at this length (Section 3.6.3); this validation establishes the estimator's sensitivity to ordering, separately from any claim about the data itself.

Paired comparisons were computed as:

$$\Delta_H = H_{Subject} - H_{PCS}$$

Because both streams are derived from the same call, the finite-sample offset cancels identically in ΔH: any shift in ΔH reflects an asymmetry in ordering structure between the streams rather than an artifact of the estimator. Primary experimental contrasts were conducted on paired differences unless otherwise noted.

2.7 Statistical Framework

Hit rates and Hurst exponents have different distributional properties and were analyzed accordingly.

2.7.1 Hit rate

Each block's hit count follows a Binomial (150, p) distribution. Hit-rate effects were analyzed using a Bayesian hierarchical model. The model included fixed effects for condition (human and AI relative to baseline) and within-session block position, plus random intercepts for each session to account for clustering. Priors were weakly informative: Normal(0, 1.5) on the global intercept, Normal(0, 0.5) on condition and block-order coefficients, HalfNormal(0.5) on the session variance scale, and Normal(0, 1) on per-session offsets. The model was implemented in

PyMC with posterior summarization via ArviZ. Posterior samples were drawn using NUTS (4 chains, 2,000 draws per chain). Results are reported as posterior means, 95% highest density intervals (HDIs), and directional posterior probabilities (Section 3.5).

2.7.2 Hurst delta (ΔH)

The sampling distribution of the single-scale R/S estimator at $n = 150$ is not well approximated by normal theory and exhibits finite-sample bias and elevated variance. Inference therefore relied on distribution-free methods: two-sample Kolmogorov–Smirnov tests for distributional comparisons and permutation tests (5,000–20,000 iterations) for label-swap, shuffle, and run-length analyses. Where t-tests are reported, they are provided for reference only; permutation-based p-values are primary.

2.7.3 Interpretation of p-values

Given the exploratory nature of the subgroup and diagnostic analyses, p-values are reported as continuous measures of compatibility with the relevant null hypothesis rather than as binary significance decisions, following recommendations against dichotomized inference (e.g., Wasserstein & Lazar, 2016, The ASA Statement on p-values). No family-wise multiplicity correction was applied to exploratory tests; where a corrected threshold would alter the qualitative reading of a result, this is stated explicitly in the text. Confirmatory claims are reserved for pre-registered replication.

2.8 Unit of Analysis

Block-level analyses describe structural differences within the observed data. Each QRNG call is physically independent, and the task contains no adaptive stimulus–response channel. However, blocks contributed by the same participant are not statistically independent.

We therefore distinguish between two questions: **within-condition structure (**does the subject stream differ from its paired control stream? Block-level statistics describe this.) and population-level generalizability (does the pattern replicate across a new, independent sample?). The first is addressable within this pilot. The second is not: with three contributors in the high-engagement subgroup, between-person variance is not adequately captured, and the individual-level outcome statistic carries an inherent noise floor that prevents reliable inference about specific participants. Population-level replication through pooled block distributions is the question Exp5 is designed to answer.

For the 5+ subgroup (3 participants, ~840 blocks), participant-level inference is limited. Leave-one-out analyses are reported in Section 3.

2.9 Exploratory Subgroup: High-Engagement Participants

Because each session was very short (30 trials, under three minutes total), we examined whether the number of sessions completed made a difference. It is possible that a single brief session is not enough for any measurable pattern to emerge.

Three participants completed five or more sessions. Together, they contributed about 840 blocks. Examined separately, this subgroup showed stronger patterns than the full sample. Because of that, we treat this as a case series of high-engagement participants for exploratory analysis.

The block count supports within-dataset statistical description, but with only three contributors, the findings cannot be generalized across observers. All results from this subgroup are exploratory. They need to be tested again in a larger, pre-registered study focused specifically on repeat participants.

There is also an important confound: session count overlaps with recruitment source. The 5+ session participants were mostly remote viewing practitioners invited to return, while most single-session participants came from a survey platform without prior interest in psi-related topics. This means the pattern cannot be interpreted as a simple learning or dose-response effect. It could reflect population differences, self-selection, repeated exposure, or some combination.

To separate those possibilities, a future study would need to recruit both populations intentionally and track session count within each group. For now, this subgroup analysis should be understood as exploratory pattern detection and not as proof of training effects or population-level differences.

2.10 Computational Environment

All analyses were conducted using Python (NumPy, SciPy, pandas). The full analysis pipeline and frozen dataset are available in the public repository (link provided in Data Availability). The analysis pipeline (Notebooks 1 and 2) extracted hit counts and raw bit arrays from the frozen dataset; additional metrics stored by the experiment application (including within-block lag-1 autocorrelation and cumulative range) were not extracted or analyzed and are documented in the repository variable schema.

3. Results

3.1 Design Validation: PCS Randomness

The Paired Control Stream (PCS) serves as the internal baseline for the experimental design. For the paired-delta architecture to be valid, the PCS must behave as an unbiased random reference across all experimental conditions.

Analysis confirmed that the PCS hit rate remained centered at 0.500, with no significant differences observed between conditions (One-way ANOVA; $p > .50$). Similarly, the PCS Hurst exponent (H) remained stable at approximately 0.53 across all conditions $(p > 0.50)$. The slight elevation of the Hurst estimate above 0.5 is consistent with the known positive finite-sample bias of R/S analysis for $(n = 150)$. Because the paired-delta metric subtracts simultaneous streams generated within the same QRNG call, this constant estimator bias is eliminated in all paired comparisons. The overlapping PCS distributions across conditions are shown in Figure 2.

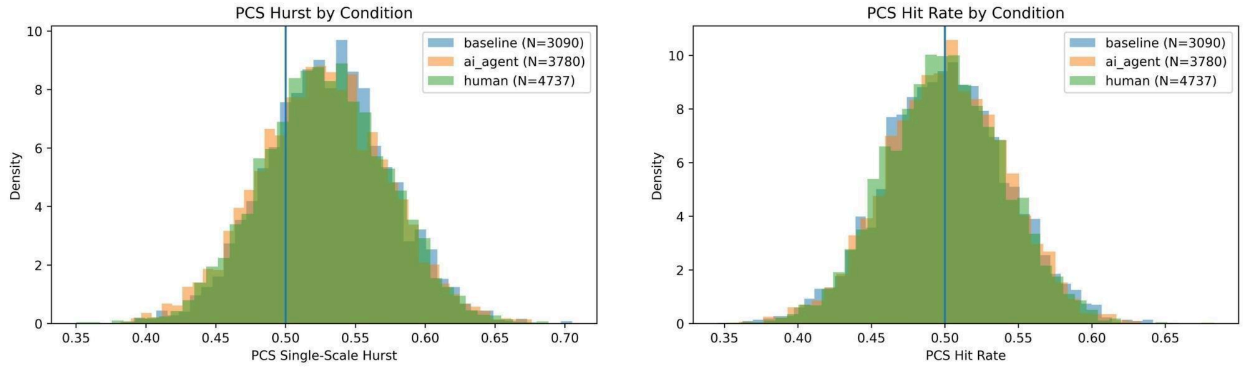

Figure 2. PCS Validation. Distributions of PCS hit rate and single-scale Hurst exponent across human, AI, and baseline conditions. No significant condition effects were detected (one-way ANOVA, $p > 0.50$), indicating that the Paired Control Stream (PCS) functions as a stable, condition-invariant reference.

These analyses establish that the PCS behaves as a stable control stream in isolation. However, PCS stability alone does not guarantee validity of the paired-delta null distribution if within-call coupling were present. We therefore next evaluate the paired null directly through within-call versus cross-call comparisons.

3.2 Validation of the Paired-Delta Null Distribution

A potential concern with the paired-delta design is circularity: if the two halves of a single QRNG call share common-mode structure, the null distribution of the paired difference ΔM cannot be assumed from the independence of the halves, and validating the PCS in isolation (Section 3.1) would not establish the validity of the delta metric itself. We therefore characterized the null distribution of the paired delta directly, using baseline-condition blocks only ($n = 3,090$; no human or AI observer present), so that these tests reflect pure hardware behavior.

3.2.1 Common-mode correlation

If consecutive halves of a single call shared hardware-induced structure, $H_{Subject}$ and H_{PCS} would be positively correlated across blocks. No such correlation was detected: for the Hurst exponent, Pearson r = +0.0037 ($p = 0.84$; Spearman $\rho = + 0.0027$; permutation $p = 0.85$, 2,000 shuffles), and for hit rate, $r = + 0.0149$ ($p = 0.41$). Results were equivalent when computed across all 11,607 blocks (Hurst $r = + 0.0085$, $p = 0.36$).

3.2.2 Empirical null of ΔH

The observed baseline ΔH distribution was centered at zero (mean = −0.0017, t-test vs. 0: $p = 0.13$), approximately normal (Shapiro–Wilk $p = 0.20$; D'Agostino $p = 0.20$), and showed negligible skew (−0.05) and excess kurtosis (−0.11). The baseline Δrate distribution was likewise centered (mean = +0.0002, $p = 0.85$).

3.2.3 Within-call vs. cross-call comparison

To test directly whether within-call pairing alters the null, the observed within-call ΔH distribution was compared to a cross-call null constructed by differencing $H_{Subject}[i] - H_{PCS}[j]$ across randomly permuted block indices (500 permutations), which destroys any within-call coupling. The two distributions were statistically indistinguishable in both shape (KS $D = 0.0095$, $p = 0.94$) and variance (Levene $p = 0.78$; variance ratio 0.9966, implying within-call $r = + 0.0034$). The same held for $\Delta rate$ (KS $p = 0.80$; Levene $p = 0.52$).

3.2.4 Interpretation

No common-mode structure between paired halves was detected in this hardware. The null distribution of the paired delta is therefore empirically validated: it matches the distribution expected under independence of the two halves, and inference on ΔM is not compromised by hidden within-call coupling.

This also clarifies the architecture's role in this study. The paired subtraction was empirically inert as there was no artifactual structure to remove. The PCS functioned as a validated control against a class of artifacts that did not materialize, at the cost of a √2 increase in the standard deviation of the delta metric. Whether the architecture performs its intended function when common-mode artifacts are present is addressed by the positive-control injection test (Section 3.3).

3.3 Artifact Injection Test (Positive Control)

Because no common-mode structure was detected in the observed hardware (Section 3.2), the cancellation property of the paired-delta architecture could not be demonstrated on naturally occurring artifacts. We therefore performed a positive-control test: synthetic common-mode artifacts of known magnitude were injected into baseline-condition data ($n = 3,090\ blocks$), which Sections 3.1–3.2 establish behaves as expected under independence (centered hit rate, stable Hurst exponent, and no common-mode correlation); providing a clean reference against which to measure injected-artifact cancellation. The response of single-stream versus paired-delta analyses was compared across a dose–response sweep.

3.3.1 Shared hit-rate bias (bit-level)

A per-call bias *a* ~ N(0, σ) was applied identically to both halves at the raw bit level ($\sigma = 0.02\ -.20$). Single-stream distortion of the Subject hit-rate distribution grew monotonically with σ (KS D up to 0.227; Standard Deviation (*SD*) inflation up to 2.57×). The paired delta remained nearly unaffected (KS D ≤ 0.014 at all levels). Cancellation at $\sigma \geq 0.05$ ranged from 70.9% to 94.4% (mean 84.2%), increasing with artifact magnitude. Cancellation is incomplete for bit-level bias because the injection acts multiplicatively on bit probabilities; the residual in the paired delta scales as $a\ x\ \Delta rate$, which is small when $\Delta rate \approx 0$.

3.3.2 Shared Hurst offset (metric-level)

A per-call offset $b \sim N\,(0,\ \sigma_H)$ was added to both $H_{Subject}$ and H_{PCS} $(\sigma_H = 0.01\text{–}0.055)$. This is the additive case corresponding to the standard model of hardware drift. Single-stream distortion again grew with σ_H (KS D up to 0.110), while the paired delta was unchanged at all levels (KS D = 0.0003; cancellation 97.8–99.7%, mean 99.1%), as expected from

$$(H_{Subject} + b) - (H_{PCS} + b) = H_{Subject} - H_{PCS}$$

3.3.3 Stream-specific control

To confirm the architecture does not erase genuine stream-specific effects, the same injections were applied to the Subject half only. The paired delta retained the injected effect at all tested magnitudes (survival 61–77% of single-stream KS distortion for σ ≥ 0.05). Survival below 100% is expected from variance scaling alone: the delta metric has √2 ≈ 1.41 × the baseline *SD* of a single stream, so an identical absolute effect produces a proportionally smaller KS distance (predicted survival ≈ 71%). The lowest injection level $(\sigma = 0.02)$ lies at the identifiability boundary of the survival statistic, where the injected effect is comparable to finite-sample variability, and is therefore not included in survival summaries.

3.3.4 Interpretation

The architecture behaves as a selective filter: common-mode structure is cancelled (97.8–99.7% for additive metric-level artifacts; 71–94% for bit-level bias at moderate-to-large magnitudes) while stream-specific effects survive at the level predicted by variance scaling.

Together, Sections 3.2 and 3.3 define the PCS's role. Its $\sqrt{2}$ variance cost is known, bounded, and recoverable through additional blocks. An undetected common-mode artifact, by contrast, produces a bias that additional data only entrenches. The trade is therefore asymmetric in the architecture's favour even when, as here, no artifact materializes.

More fundamentally, the conclusion that this hardware was clean is itself a product of the paired design: the within-call versus cross-call comparison is only possible because every call carries its own simultaneous control. Without the PCS, hardware stability would rest on provider documentation; with it, within-call independence was measured under the study's actual operating conditions — three public-API providers, exchanged mid-study, without per-block provider logging. These analyses do not prove the absence of hardware artifacts; they place empirical bounds on detectable within-call coupling under the conditions sampled.

3.4 QRNG Hardware Validation

To determine if observed deviations could be attributed to hardware instability or environmental artifacts, we performed a comprehensive audit analysis using 1,891 independent 1000-bit audit sequences collected between task blocks. These audits serve as condition-blind "probes" of the hardware state. The results across eight distinct validation tests confirm that the generator remained stable and unbiased (Table 1) throughout the study.

Table 1. QRNG Hardware Validation and Stability Audit Results

Validation Test	Metric	Statistical Result	Outcome
Condition Invariance	Hurst (*H*)	$F = 1.416,\ p = 0.24$	Stable across groups
NIST Randomness Suite	Pass Rate	$\chi^2 = 0.199$	97.7% Pass (Random)
Hardware Drift	Correlation (r)	$r = -0.0048,\ p = 0.84$	No block-to-block clumping
Chronological Drift	Spearman ρ	$\rho = -0.0347,\ p = 0.13$	No long-term decay

Warm-up Effect	$t - test$ (Early/Late)	$t = 0.684,\ p = 0.49$	Stable from start to finish
Entropy Stability	Shannon Entropy	$p = 0.94$	Consistent bit-depth
Variance Stability	Levene's (W)	$W = 1.431,\ p = 0.23$	No noise inflation
Periodicity	FFT Power Ratio	Ratio = 3.17 (Expected < 6.9)	No rhythmic cycles

[a] See Table 2 for the expanded NIST sub-test results

Table 2. Detailed NIST SP 800-22 Test Results by Experimental Condition

NIST Statistical Test	Human (Pass %)	AI Agent (Pass %)	Baseline (Pass %)	Mean p-value

Frequency (Monobit)	98.2%	98.7%	97.7%	0.4915
Runs	100.0%	100.0%	100.0%	0.6138
Longest Run of Ones	99.1%	98.2%	98.9%	0.4986
Overall Pass Rate	97.8%	97.6%	97.5%	—

Note: Chi-square test for pass rate uniformity across conditions: $\chi^2 = 0.199, p = 0.91$.

Across all 1,891 sequences, 44 (2.33%) failed at least one NIST sub-test. Beyond the condition-invariance already shown, these failures showed no clustering by session (each occurred in a distinct session), by time ($\chi^2 = 1.14$, $p = 0.89$ across months), or by provider, with a weak, non-significant elevation for the LFDR backup source (4.05% vs. 2.21%; Fisher's exact $p = 0.24$) noted given its small audit sample. These validation results indicate that audit failures occurred at a low rate with no systematic structure, consistent with the QRNG behaving as a stable randomness source throughout the study period.

3.5 Primary Outcome: Hit Rate Analysis

To evaluate the presence of an anomalous effect, we employed Bayesian parameter estimation to calculate the difference in success probability Δp between the Subject and PCS streams.

At the population level, pooling all human blocks against baseline showed no significant difference in paired hit-rate delta ($p = 0.76$). The Bayesian model therefore focused on the pre-specified high-engagement subgroup. For the experienced human cohort, the model yielded a posterior mean of $\Delta p = + 0.0021$ (95% HDI: [-0.0010, +0.0053]). While the 95% highest density interval (HDI) included zero, the directional probability was notable, with an 89.8% probability that the effect was positive ($Pr(\Delta p > 0) = 89.8\%$).

In contrast, the AI condition was statistically indistinguishable from the baseline, yielding an estimated Δp = + 0.0001 (effectively zero). Direct comparison of the two groups indicated an 89.2% probability that the human performance exceeded that of the AI agent ($Pr(human > AI) = 89.2\%$).

While the hit-rate analysis reveals a directional trend favoring experienced human participants over the AI control ($Pr(human > AI) = .892$), it does not provide evidence of a significant mean shift; the 95% HDI for the human-baseline difference includes zero.

A pooled analysis of all 1,741,050 bits confirmed no significant deviation from the theoretical mean ($p = 0.827$), with a nominal human-to-baseline difference of only +206 ppm, supporting the necessity of moving beyond mean-based metrics to analyze temporal structure.

Two moderators commonly examined in this literature were pre-collected and assessed as exploratory checks. For psi-related belief, the subject-stream hit rate was marginally higher in believers than skeptics, consistent with the classical direction. However, the paired-delta hit rate reversed this, and Hurst delta did not differ (all $p > 0.28$). For meditation, the paired-delta hit rate was marginally higher in regular meditators, while Hurst delta ran in the opposite direction ($p = 0.10$); the hit-rate contrast was far from significant ($p = 0.87$).

In both cases the direction was inconsistent across metrics, and no comparison approached significance. Given the short protocol and the design's limited power for small effects (Section 5.7), these analyses cannot distinguish a true null from an effect below the detectable threshold. No moderator effect was detected, and these checks are peripheral to the methodological aim of the study.

3.6 Exploratory Ordering Analysis: Hurst Temporal Structure

3.6.1 Population-level null

When all human blocks were pooled and compared to baseline on Hurst delta (ΔH), the difference did not reach conventional significance (KS p = 0.052; Welch t

= +1.94), reflecting the fact that single-session participants are statistically indistinguishable from baseline.

3.6.2 Session-Count Pattern (Exploratory)

Hurst delta (ΔH) increased with session count:

- 1 session: mean Δ ≈ 0.000 (baseline-like), KS p = 0.565
- 2–4 sessions: mean Δ = +0.003, KS p = 0.244
- 5+ sessions: mean Δ = +0.004, KS p = 0.040, d = +0.095
- AI agent: mean Δ = +0.001 (not significant)

The separation was largest in the 5+ session subgroup (KS p = 0.040); see Section 2.7.3 on the continuous interpretation of these p-values.

The session-count pattern in mean ΔH is illustrated in Figure 3.

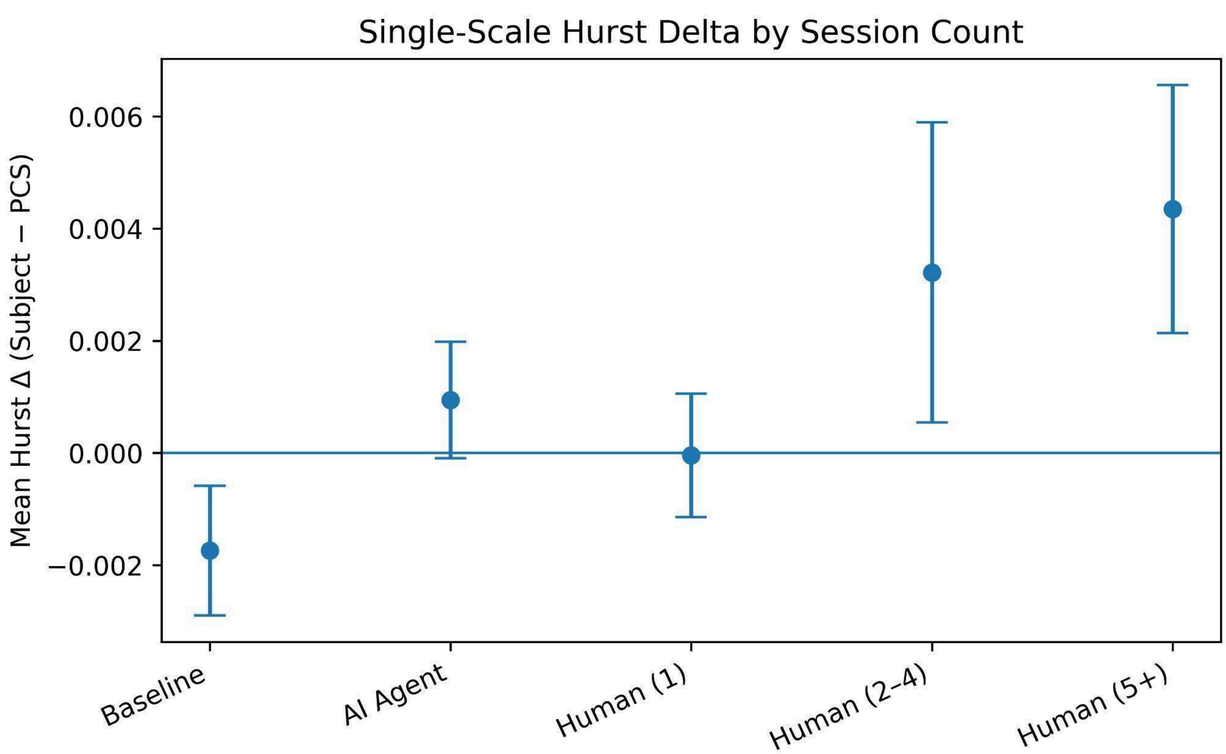

Figure 3. Mean single-scale ΔH (Subject − PCS) by session count. Error bars show ±1 SE. Only the ≥5 session subgroup differs significantly from baseline at the distributional level (Kolmogorov–Smirnov test, p = 0.040).

3.6.3 Shuffle Test: Metric Sensitivity to Within-Block Ordering

Because R is computed on the cumulative-deviation path rather than on the raw bit values, both R and the resulting H depend on the order in which bits appear within a block (Section 1.3, Section 2.6.2). Shuffling the bits within a block traces a different cumulative path, yields a different R, and therefore a different H. This is a property of the estimator, not an empirical claim about the data: the single-scale R/S statistic is order-dependent by construction.

The shuffle test characterizes how the estimator responds to within-block temporal ordering at the 150-bit block length used here. Subject bits were shuffled within progressively larger windows (W = 1, 3, 5, 10, 25, 50, 150 bits). After each shuffle, $H_{Subject}$ was recomputed and the paired delta ($\Delta = H_{Subject} - H_{PCS}$) was recalculated while leaving the PCS unchanged. Shuffling at $W = n = 150$ constitutes a full-block randomisation that destroys all within-block ordering while preserving overall bit composition. We report the mean *KS* distance *(*KS D*)* between the subgroup and baseline ΔH distributions as a function of W. The procedure measures the degree of the estimator's ordering sensitivity at this block length; it is not a hypothesis test about the significance of any group difference.

For the Human 5+ subgroup, the KS D between the subgroup and baseline ΔH distributions varied with shuffle window size and was lowest under full-block shuffle (W = 150), indicating that the single-scale R/S estimator is sensitive to within-block ordering at this block length. For the AI condition, KS D remained near its unshuffled level across all window sizes, showing no comparable ordering sensitivity. The two curves are shown in Figure 4.

This contrast is consistent with the property the metric was selected for: its output reflects within-block temporal organisation rather than bit composition alone, and this sensitivity is present in the human subgroup's data and absent in the AI condition. No inference about the significance of any group difference is drawn from the shuffle procedure; it serves only to confirm that the estimator is an appropriate instrument for ordering structure at $n = 150$.

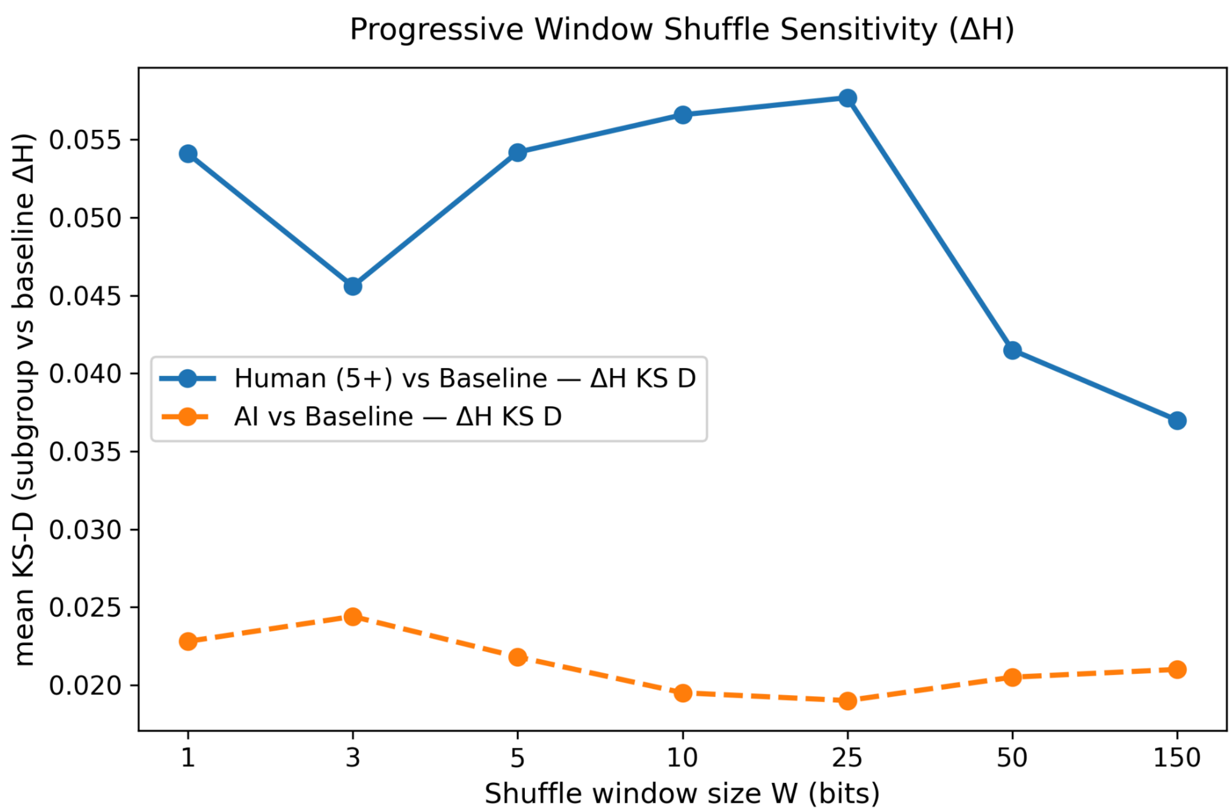

Figure 4. Estimator sensitivity to within-block ordering. Mean KS distance between subgroup and baseline ΔH distributions, plotted against shuffle window size W (200 shuffles per window; windows shown on an evenly-spaced categorical axis). For the Human 5+ subgroup, KS D varies with W and is lowest under full-block shuffle (W = 150). The AI condition shows no comparable variation across window sizes. The figure characterises how the single-scale R/S estimator responds to ordering at $n = 150$; it is not a test of the significance of any group difference.

These results characterise the single-scale estimator's response to within-block ordering at $n = 150$: KS D varies with shuffle window in the human subgroup's data and is lowest under full-block randomisation, while the AI condition shows no comparable variation. The procedure confirms the estimator is sensitive to ordering structure at this block length; it does not, on its own, establish that the subgroup difference is ordering-dependent.

3.6.4 Label swap test: sensitivity to stream assignment

To test whether the Hurst delta (ΔH) difference depends on subject vs PCS assignment, the labels were randomly swapped within blocks (50% probability per block; 5,000 permutations per cohort). If the separation reflects symmetric hardware behavior, swapping labels should have no effect. If it depends on stream assignment, swapping should eliminate the difference.

Table 3

Results:	
Human (1 session):	$p(swap) = 0.287$
Human (2-4 sess):	$p(swap) = 0.085$
Human (5+ sess):	$p(swap) = 0.014$
AI Agent:	$p(swap) = 0.086$

Table 3. Label-swap permutation test results by cohort.

Permutation p-values for the label-swap test on Hurst delta (ΔH); labels swapped within blocks at 50% probability, 5,000 permutations per cohort. Interpretation is given in Section 3.6.4.

These p-values are reported as continuous measures of evidence against the swap-invariant null, not as dichotomous classifications. The 5+ session subgroup shows the strongest evidence of assignment sensitivity, with the 2–4 session and AI cohorts intermediate and the 1-session cohort showing none.

No formal multiplicity correction was applied across the exploratory tests in Sections 3.6.2–3.6.4 (session-count comparisons, shuffle tests, and label-swap tests constitute a family of at least eight related hypothesis tests). Under a Holm–Bonferroni correction across this family, no individual test, including the 5+ subgroup swap result ($p = 0.014$), would remain significant at the conventional family-wise α = 0.05. We therefore do not claim that label dependence is established. What can be stated is descriptive and convergent: in this dataset, the 5+ subgroup is the only cohort in which the ΔH separation is jointly sensitive to within-block ordering (Section 3.6.3) and to stream assignment, and the direction of both sensitivities is consistent across all three contributing participants. Whether this convergent pattern reflects an observer-coupled effect or a chance configuration among many exploratory comparisons is precisely the question a pre-registered replication is required to answer.

Because this subgroup consisted of only three participants and was identified following null results in the full sample, these findings should be interpreted as exploratory rather than confirmatory. The present analyses describe patterns within the observed data but do not establish generalizability and require replication under pre-registered conditions.

3.6.5 Direct-Stream / Paired-Delta Convergence and the Four-Step Diagnostic Chain

The paired-delta architecture is designed to yield two complementary analyses from the same data: the direct single-stream comparison ($H_{Subject}$, Human vs Baseline) and the paired-delta comparison ($\Delta H = H_{Subject} - H_{PCS}$, Human vs Baseline). These address structurally different questions (the direct analysis asks whether the subject stream differs from baseline; the paired delta asks whether that difference is stream-specific rather than shared across the call) and they have different statistical properties. The direct analysis carries greater sensitivity (no √2 variance penalty) when hardware is clean; the paired delta carries greater validity under artifact conditions. When both analyses agree in direction, and when the label-swap result (Section 3.6.4) is also significant, the three results form a four-step chain of evidence that characterizes the nature of the signal rather than merely its existence. Steps 1–3 are the confirmatory evidential core; Step 4 is a mechanistic exploration. The chain below is presented to demonstrate what the paired-delta architecture can do. It can discriminate among competing explanations for an observed pattern. It cannot claim that an observer-linked effect is established in this dataset. Each step is a diagnostic the design makes available that a single-channel study could not run. Whether the pattern reflects a genuine effect is a question for the pre-registered replication.

3.6.5.1 Step 1, Signal existence

For the 5+ session subgroup, the direct $H_{Subject}$ vs Baseline comparison yields KS $D = 0.0575$, $p = 0.0243$. A signal is present in the Subject stream (Human 5+).

3.6.5.2 Step 2, Stream-specificity

The paired ΔH vs Baseline comparison yields KS $D = 0.0541$, $p = 0.0398$. Both analyses clear $p < 0.05$ in the same direction, and the direct analysis yields the stronger result, consistent with the theoretical prediction for clean-hardware conditions, in which the direct analysis retains full power while the paired delta pays the √2 variance cost. More importantly, the signal does not cancel in the paired delta. If the elevated $H_{Subject}$ were produced by a symmetric hardware drift affecting both halves of each call equally, the drift would raise H_{PCS} by the same amount, and the paired difference would approach zero. It does not. The signal is stream-specific.

3.6.5.3 Step 3, Label-dependence

The label-swap test (Section 3.6.4) yields $p_{swap} = 0.014$. Randomly reassigning which stream is called "Subject" and which "PCS" degrades the delta. The signal tracks the label, not just the underlying quantum draw. A hardware artifact that happened asymmetrically to favor one half of a call would be label-independent (the bits don't change when you rename them); label-dependence implies the structure is genuinely linked to the assignment. Steps 1 through 3 constitute the core evidential chain: a signal exists (Step 1), it survives the simultaneous hardware control (Step 2), and it tracks the stream label rather than the underlying quantum draw (Step 3). Step 4 is an additional mechanistic observation specific to this pilot dataset. It describes why the delta is large and label-sensitive here, but it is not a required component of the chain. A replication that supports Steps 1–3 without Step 4 would constitute meaningful evidence of an observer-linked, artifact-controlled signal; Step 4's absence would be informative about mechanism, not evidence against the effect's existence.

3.6.5.4 Step 4, Bidirectional mechanism (exploratory)

To understand why the delta is label-sensitive, the individual stream distributions were examined separately for each cohort. For the 5+ subgroup: the Subject stream (Human 5+) trends above baseline (t = +1.770, p = 0.077) while the PCS (Human 5+) trends below baseline (t = −1.668, p = 0.096). The two streams from the same 301-bit call are moving in opposite directions relative to baseline. This bidirectional divergence is the mechanism underlying Steps 2 and 3: the streams compound the delta (both contributing to its magnitude in the same direction by diverging in opposite directions), and when labels are swapped, the elevated stream sometimes moves to the PCS position and the depressed stream to the

Subject position, partially canceling the effect, which is precisely what p_{swap} detects. Table 4 below shows this pattern across cohorts:

Table 4: Stream-level directional pattern and label-swap sensitivity by cohort

Cohort	p_t	p_swap	Subject	PCS
Human 1 session	0.287	0.287	↑ ns	↔ ns
Human 2–4 sessions	0.090	0.085	↑ marginal	↔ ns
Human 5+ sessions	0.015	0.014	↑ marginal	↓ marginal
AI agent	0.084	0.086	↔ ns	↔ ns

The 5+ subgroup is the only cohort in which both streams are moving, in opposite directions, and in which both p_t and p_{swap} fall below 0.05. In all other cohorts, neither stream shows a consistent directional pattern, and neither test clears $p < 0.05$.

What this chain rules out: Symmetric common-mode drift (temperature, line noise, hardware) would move both streams in the same direction, elevating $H_{Subject}$ and H_{PCS} together, producing near-zero ΔH and a label-swap null near $p = 0.5$. That is not the pattern observed. A story in which the 5+ participants happened to be allocated a temporally-ordered block of QRNG output (without any stream-specific mechanism) would also elevate both streams together, again inconsistent with the opposite-direction divergence. The bidirectional pattern is structurally harder to explain by any mechanism that acts equally on both halves of a call.

What this chain does not establish: Whether the bidirectional pattern reflects anything real or is a chance configuration across a handful of tests applied to three participants. The 840 blocks support valid statistics at the block level, but $N = 3$ participants means participant-level variance is not adequately captured. The leave-one-out analyses (Section 3.7.5) are the appropriate next check; all three participants contribute in the same positive direction, which provides some robustness, the chain as a whole is descriptive of this dataset and Steps 1-3 require replication at scale before any broader interpretive claims can be made.

The direct-stream analysis reported in Steps 1–2 was not pre-specified as a primary outcome in the pilot (the paired ΔH was); it is included here as an architecturally motivated exploratory companion, the natural complement the design was built to provide, rather than as an independent confirmatory test. Steps 1–3 are pre-specified as the confirmatory diagnostic sequence in the Exp5 pre-registration (Section 4.4); Step 4 is pre-specified as a mechanistic exploration, examined conditionally if Steps 1–3 are supported.

3.6.6 Run-length validation

To verify that elevated Hurst values correspond to measurable ordering differences, blocks in the extreme tails of the baseline Hurst distribution (1st and 99th percentiles) were analyzed using direct run-length metrics computed from raw bit sequences (20,000 permutation tests).

Compared to typical blocks, extreme-Hurst blocks showed longer mean run lengths, larger maximum runs, more long-run episodes and fewer total runs. All four reached $p < 0.001$. Run-length entropy did not reach significance, likely reflecting mixed long- and short-run structure within extreme blocks. This is consistent with elevated Hurst values corresponding to measurable differences in bit ordering, rather than numerical instability.

Importantly, the proportion of extreme-Hurst blocks was similar across conditions (baseline 2.01%, AI 1.61%, human 1.69%), indicating that the metric does not disproportionately flag any condition.

3.6.7 Interpretation

Taken together, these diagnostics converge on a single subgroup: the high-engagement cohort is the only group in which the ΔH distribution differs from baseline, the difference is sensitive to within-block ordering, and it depends on stream assignment, while composition (entropy), variance, session structure, and hardware audits show no corresponding differences. This convergence rules out several specific artifact-based explanations: the observed pattern is inconsistent with composition bias, noise injection, symmetric hardware drift, and session batching. It cannot, however, exclude a chance configuration across multiple exploratory tests. Distinguishing structure from chance is the specific task of the pre-registered replication.

3.7 Additional Robustness Checks

3.7.1 Within-Session Drift

No early–late differences in ΔH were observed within sessions. For 5+ participants, early and late halves did not differ significantly $(+\ 0.00586\ vs\ +\ 0.00284; p\ =\ 0.495)$. This suggests the effect is not attributable to within-session warm-up or fatigue.

3.7.2 Entropy

Shannon entropy of subject bits did not differ across conditions (Human vs baseline $p\ =\ 0.829$; Human 5+ vs baseline $p\ =\ 0.167$). Entropy was uncorrelated with $\Delta H(r \approx 0,\ p\ =\ 0.707)$. The observed difference therefore reflects ordering structure rather than changes in bit frequency.

3.7.3 Variance equality

The spread of the Hurst delta did not differ across conditions (Levene's test, all human strata vs baseline $p\ >\ 0.76$; permutation variance ratios near 1.0, all $p\ >\ 0.47$). The small mean shift in the experienced subgroup is therefore not accompanied by increased variability, distinguishing a directed-shift pattern from random noise injection.

3.7.4 Session Structure / Batching

A matched resampling procedure (5,000 iterations) replicated the session structure of the 5+ subgroup using baseline blocks. The observed aggregate mean ΔH was unlikely under baseline-only structure (one-sided $p = 0.002$), and the distributional shape differed from session-matched resamples (KS $p = 0.011$). Together with the shuffle and label-swap tests, this indicates the effect cannot be explained solely by session batching or block grouping.

3.7.5 Leave-One-Out Sensitivity

The 5+ subgroup consists of three participants. Removing each in turn weakens statistical significance, and removal of the strongest individual contributor renders the KS test non-significant. All three participants contribute in the same positive direction.

3.7.6 First vs Repeat Participation

Single-session participants do not differ from baseline (KS $p = 0.565$). Repeat participants show positive delta values from their first session, but estimates are noisy. The present data cannot distinguish between self-selection and exposure effects.

Taken together, the shuffle dependence, label-swap sensitivity, PCS invariance, and audit stability are not consistent with simple symmetric hardware drift alone.

Table 5 reports this sensitivity across cutoffs and with/without the PI; effect size (Cohen's d) does not decrease when the PI is excluded, indicating the loss of significance reflects reduced power (n=3→2) rather than a smaller effect.

Table 5 Hurst-delta sensitivity across session-count thresholds and PI inclusion.

Subgroup	***N* participants**	***N* blocks**	**Mean Delta*H***	**KS *D***	**KS *p***	***t***	***t p***	**Cohen's *d***
1 session (reference)	111	3327	0	0.0195	0.5648	+1.064	0.2872	+0.027
2+ sessions (incl. PI)	10	1410	+0.0039	0.0426	0.0577	+2.735	0.0063	+0.088
2+ sessions (excl. PI)	9	870	+0.0051	0.0437	0.1438	+2.777	0.0055	+0.107
3+ sessions (incl. PI)	6	1170	+0.0034	0.0464	0.0496	+2.346	0.019	+0.081
3+ sessions (excl. PI)	5	630	+0.0047	0.0552	0.0789	+2.288	0.0222	+0.100
5+ sessions (incl. PI)	3	840	+0.0043	0.0541	0.0398	+2.441	0.0147	+0.095
5+ sessions (excl. PI)	2	300	+0.0087	0.0787	0.064	+2.682	0.0074	+0.162

Effect size (Cohen's $d \approx$ 0.08–0.16) is stable across all session-count thresholds; the 5+ result is not an artifact of this specific cutpoint.

4. Discussion

4.1 Architectural Contribution

Section 1.1 identified two distinct problems in this literature. The paired-delta architecture directly addresses Problem 2: within-study artifact ambiguity. By generating Subject and PCS streams within the same QRNG call, differing only by a quantum-random label assignment, the design controls for common-mode hardware drift, timing artifacts, and environmental variation at the level of each call. The subtraction is implemented at the data-generation level rather than introduced as a post-hoc adjustment. The logic is analogous to differential signaling in electronics: two streams share the same underlying physical pathway, and subtraction removes variation that affects both equally. Any remaining difference must arise from asymmetries between the labeled streams.

The architecture does not resolve Problem 1 (cross-study credibility). Publication bias, selective reporting, and analytic flexibility are process-level problems that require process-level remedies including pre-registration, locked datasets, and transparent reporting that are applied before outcomes are known. This paper addresses Problem 1 partially, through in-code metric pre-specification and a cryptographic data freeze; the complete remedy is the pre-registered replication this pilot motivates (Section 4.4). The two responses are complementary rather than equivalent: the architecture addresses within-study validity; the replication addresses cross-study credibility. Neither substitutes for the other.

In this dataset the cancellation mechanism was not exercised by real artifacts as no common-mode structure was detected in the hardware (Section 3.2). The architecture's value here is the validated null distribution it provides and the insurance it offers against provider or session-level artifacts in future deployments. That the hardware was clean is itself a product of the paired design: the within-call versus cross-call comparison in Section 3.2 is only possible because every call carries its own simultaneous control.

4.2 What the Pilot Data Suggest

This study was designed primarily to develop and evaluate the paired-delta architecture rather than test for observer-linked deviations from chance. Detection of ordering-sensitive structure was a secondary exploratory objective, and the dataset was not powered for confirmatory inference.

The protocol was intentionally brief (30 trials per session; 150-bit blocks), prioritizing accessibility over statistical resolution. Under these constraints, hit-rate differences were weak, while a modest distributional shift in Hurst delta (ΔH) was observed in a higher-engagement subgroup.

Because this subgroup consisted of only three participants and was identified following null results in the full sample, these findings should be interpreted as exploratory and require replication under pre-registered conditions.

4.3 The Dual-Analysis Property and What the Pilot's Chain of Evidence Shows

The validation results in Section 3 reframe what this architecture provides beyond artifact control. Because the Subject and PCS streams derive from the same QRNG call (Section 1.2) and the two halves are empirically independent (Section 3.2), every call yields two complementary analyses for the cost of one: the direct single-stream comparison (full statistical power, no variance penalty, valid when hardware is clean) and the paired-delta comparison (artifact-immune by construction, at the cost of a √2 variance increase). These are not independent tests (they share the same underlying data), but they answer structurally different questions and have different operating characteristics. Together they enable a chain of inference that goes beyond asking whether a difference exists to characterizing what kind of difference it is.

The pilot's 5+ session subgroup demonstrates this chain illustratively (Section 3.6.5):

Step 1 (existence): The direct $H_{Subject}$ vs Baseline comparison yields KS $p = 0.0243$, a signal is present.

Step 2 (stream-specificity): The paired ΔH vs Baseline comparison yields KS $p = 0.0398$, the signal survives subtraction of the simultaneous control stream, ruling out symmetric hardware drift as the sole explanation.

Step 3 (label-dependence): The label-swap test yields $p = 0.014$, scrambling which stream is called "Subject" degrades the delta, meaning the signal tracks the label assignment, not just the underlying quantum draw.

Step 4 (bidirectional mechanism): Individual stream examination shows the Subject stream (Human 5+) trending above baseline ($t = + 1.770$, $p = 0.077$) while the PCS (Human 5+) trends below $p = 0.077$($t = - 1.668$, $p = 0.096$), opposite directions from the same 301-bit call. This bidirectional divergence is the mechanism underlying Steps 2 and 3: the streams compound the delta (both contributing in the same direction by diverging oppositely), and label-swapping partially cancels this by sometimes placing the depressed stream in the Subject position.

The chain as a whole rules out symmetric common-mode explanations, hardware drift, environmental noise, or session batching, more cleanly than any single diagnostic. Common-mode mechanisms move both streams in the same direction; bidirectional divergence requires a mechanism that acts differently depending on which stream is labeled Subject. No single-channel design can demonstrate this distinction because it has nothing to compare against.

Steps 1 through 3 constitute the core evidential chain. A replication supporting all three — signal exists (Step 1), survives hardware control (Step 2), and is label-dependent (Step 3) — constitutes meaningful evidence of an observer-linked, artifact-controlled signal regardless of Step 4's outcome. Step 4 is a pilot-specific mechanistic observation: the bidirectional stream divergence that explains why the delta is large and label-sensitive in this particular dataset. In Exp5, Step 4 is examined exploratorily rather than as a necessary condition for replication. A replication where Step 4 is absent (subject elevated but PCS not depressed) would indicate the effect, if real, operates through unilateral subject-stream elevation rather than bidirectional divergence, informative for mechanism rather than evidence against existence.

In Exp5, the full chain is pre-registered as a conditional diagnostic sequence applied if and only if H0 is supported at the confirmatory threshold (Exp5 OSF preregistration, [DOI to be added upon publication]). Steps 1–3 are the evidential core; Step 4 is the mechanistic exploration. This structure allows Exp5 to confirm, modify, or falsify the pilot's mechanistic claim without the chain's exploratory component invalidating a genuine replication.

4.4 Planned Pre-Registered Replication and Permeability Framework

These design principles have been formalized in a pre-registered follow-up study (Exp5) that has been designed and registered but not yet initiated; data collection is contingent on the outcome of the present work. The planned study extends the present findings in two directions: (a) a fully confirmatory replication of the paired-delta and direct-stream signatures under locked analytic conditions, including pre-specified shuffle and label-swap validation gates; and (b) an exploratory individual-difference framework targeting permeability (a hypothesized reduced filtering at the boundary between the person and available information) operationalized through a set of individual-difference predictors analyzed separately (psi-content items — belief, self-assessed ability, and anomalous-experience breadth — together with meditation practice, the MAIA-2, the BQ-18, and, pending permission, the HSP-12 and the Revised Transliminality Scale).

The planned design isolates confirmatory inference to the replication tests (the existence and reproducibility of the architectural signature), while treating all trait-predictor analyses as exploratory, hypothesis-generating groundwork toward the future development of a dedicated permeability measure. It therefore asks first whether the architectural signature generalizes at scale, and second, exploratorily, which, if any, individual-difference channels track its magnitude.

4.5 The Single-Scale Hurst Exponent as a Metric

The single-scale Hurst estimator (implemented as hurstApprox) was specified prior to data collection as a screening instrument sensitive to within-block ordering. Its use reflects a scale-matching decision appropriate to 150-bit blocks, not a claim that ordering structure is a primary outcome.

For short sequences ($n = 150$), multi-scale regression methods partition the data into small subwindows. At this length, those subwindows are highly variable, and regression across them introduces additional estimator variance. A single-scale R/S statistic, computed across the full block, provides a lower-variance summary of within-block ordering.

For longer sequences, the situation changes. When n is substantially larger (e.g., thousands of bits), multi-scale regression becomes stable because window sizes are sufficiently large to reduce noise. In that regime, multi-scale approaches allow characterization of ordering structure across scales rather than at a single block length. Estimator choice should therefore be matched to sequence length and study design.

To verify that the Hurst estimator reflects genuine sequence structure rather than numerical artifacts, we performed run-length validation. Blocks flagged as extreme by Hurst show longer mean runs, larger maximum runs, and more long-run episodes (all $p < 0.001$ under permutation testing) in the underlying bit sequences.

For future studies, pre-registering both hit rate and a block-length–appropriate ordering metric is recommended. Hit rate measures aggregate bit frequency; the Hurst metric serves as a screening flag for ordering structure (to be confirmed or dismissed by the convergent order-based diagnostics). Together, they separate mean-level effects from ordering-based structure within the paired architecture.

4.6 Temporal Ordering vs Bit Composition

Beyond estimator choice, the pilot data highlight a broader methodological distinction between bit composition and temporal ordering.

Shannon entropy of subject bits did not differ across conditions ($p = 0.829$) and was uncorrelated with Hurst delta (ΔH) ($r \approx 0$). Human and baseline blocks contain the same proportion of 1s and 0s. Any observed difference lies in the ordering of those bits rather than in their overall frequency. Mean-based metrics such as hit rate assess composition: whether the total number of outcomes departs from expectation. Ordering metrics such as the Hurst exponent assess persistence or clustering within a sequence. These are analytically distinct properties. A sequence may match expected frequencies while exhibiting non-independent ordering.

Large-scale QRNG studies have primarily relied on mean-based analyses. Maier et al. (2018), for example, reported strong Bayesian evidence against mean deviation from chance, yet also explored whether sequential organization might follow structured temporal patterns independent of aggregate outcomes. The present work differs in that ordering structure (via a single-scale Hurst metric) was pre-specified and evaluated within a paired-control design.

In this dataset, hit-rate analyses show weak evidence of mean-level differences, while the exploratory Hurst analysis identifies a small distributional shift in paired ordering differences within a high-engagement subgroup. Because entropy and marginal frequencies remain unchanged, any replicated effect would reflect ordering differences rather than compositional bias.

This distinction is methodological rather than theoretical. It suggests that combining mean-based and ordering-based metrics within a paired-control architecture provides a broader framework for evaluating structure in short QRNG sequences while controlling for hardware-level variation.

5. Limitations

5.1 Short protocol duration

Each session contained 30 trials (150 bits per block). This short format prioritizes accessibility but limits statistical resolution and increases estimator variance relative to longer protocols.

5.2 Small high-engagement subgroup

The 5+ session subgroup consists of three participants (~840 blocks). Block-level analyses describe patterns within the observed data but cannot establish generalizability across observers; limited by the small number of contributors. Two resampling checks bound this limitation: leave-one-out analysis shows that removing any single participant weakens statistical significance, indicating sensitivity to participant composition; and clustered bootstrap resampling (resampling participants, then blocks within participant) preserves the positive direction of the effect but widens confidence intervals substantially relative to block-level resampling. Replication with a larger repeat-participant sample is therefore essential to determine generalizability.

5.3 Post-hoc subgroup analysis

The higher-session subgroup was examined following null results in the full sample and was not pre-registered. Findings from this subgroup should therefore be treated as exploratory and interpreted in conjunction with the limitations of participant count and multiplicity described below.

5.4 Multiple exploratory tests

Multiple subgroup and metric comparisons were evaluated without formal multiplicity correction. Consequently, nominal p-values should be interpreted descriptively rather than as confirmatory evidence. The qualitative effect of a Holm–Bonferroni correction on the exploratory family is reported in Section 3.6.4.

5.5 Recruitment confound

Higher session-count participants were recruited from a different source than most single-session participants. Session count is therefore confounded with recruitment channel and participant characteristics. The present data cannot distinguish between exposure effects, trait differences, or self-selection.

5.6 Experimenter participation

One member of the high-engagement subgroup was also involved in study administration. Although the paired architecture prevents mechanical influence on outcomes, inclusion of a non-blinded experimenter limits independence of that subgroup. Leave-one-out (Section 3.7.5) analyses are reported to assess sensitivity. Notably, the principal investigator's blocks show the smallest individual paired-delta among the three subgroup participants, so their inclusion lowers rather than inflates the subgroup mean; the observed pattern is not driven by the non-blinded contributor.

5.7 Statistical resolution limits

Sensitivity analysis indicates a minimum detectable effect of approximately 0.37% (block-level paired delta, 80% power) for this dataset which is roughly nineteen times larger than the mean effect sizes reported in classical RNG intention studies (e.g., Jahn et al., 1997; Bösch et al., 2006, reporting effects on the order of 0.02% deviation from chance).

This design is therefore not powered to detect classical psi-scale effects at the population level, and absence of strong mean-level effects should not be interpreted as evidence of absence for effects below this threshold. This limitation is not unique to this paradigm: Maxwell, Lau, and Howard (2015) document how underpowered single studies systematically fail to surface genuine small effects, and identify larger pooled designs and multiple replications as the recognized response. This motivates both the use of complementary order-based diagnostics and the pre-registered, higher-powered replication described in Section 4

5.8 AI control implementation

The AI agent interacted with the experimental interface via scripted automation and did not autonomously control the timing of QRNG calls. Its responses were generated statelessly and without learning across trials. As such, the AI condition represents an automated control under constrained interaction, not a fully adaptive agent with independent task control.

5.9 QRNG provider heterogeneity

QRNG calls were served through more than one hardware provider during the study period. Although the paired-delta architecture controls for within-call hardware variation and audit metrics showed stability across conditions, provider identity was not logged per block. Future studies should explicitly record and model provider source.

5.10 Execution environment variability

Participants accessed the experiment through web browsers under uncontrolled local computing environments. Network latency and device variability were not explicitly modeled. While the paired architecture mitigates common-mode timing effects within each call, future protocols should standardize or log environment characteristics.

6. Conclusion and Recommendations

This study introduces and validates a paired-delta QRNG architecture designed to control for common-mode hardware drift, timing artifacts, and environmental variation within each QRNG call. Independent audit streams, PCS stability, label-swap tests, and shuffle analyses collectively support that the architecture operated as intended under the sampled conditions.

Hit-rate analyses show weak mean-level differences. In an exploratory subgroup of three high-engagement participants, a small distributional shift in paired Hurst delta (ΔH) values was observed. Diagnostic tests showed the strongest evidence of ordering and assignment sensitivity in this subgroup, and no comparable pattern in the AI condition; given that these tests were among several exploratory comparisons, where chance alone produces occasional low p-values, the evidence is suggestive rather than confirmatory. However, the subgroup contains only three participants, and the findings require replication under pre-registered protocols with larger repeat-participant samples.

The primary contribution of this work is methodological. Simultaneous paired control streams allow separation of compositional effects from ordering structure while controlling for hardware-level variation. Ordering-based metrics can serve as screening signals that direct convergent diagnostic testing, complementing traditional mean-based measures in short-form QRNG experiments.

Progress in this area depends on designs that directly address common methodological criticisms through structural controls rather than post-hoc statistical adjustments. The paired-delta architecture is intended as a step in that direction.

Acknowledgements and Disclosures

In memory of my father, Alfred C. Rester Jr. (1940–1996), nuclear physicist and arms-control scientist, whose work ranged from the Gamma Ray Advanced

Detector to the observation of Supernova 1987A, and whose love of discovery inspired my own.

AI-Assisted Research

Large language models were used during the development of experimental code, data processing pipelines, and manuscript drafting. The study design, architectural decisions, and analytic strategy were directed by the author. All generated code and analyses were reviewed and validated against the frozen dataset prior to inclusion in the manuscript.

Data Transparency

All data collection procedures and metrics were defined in the experiment source code prior to analysis. The dataset was frozen using SHA-256 hashing before statistical evaluation. Raw data and full computational notebooks are available in the public repository referenced in the Data Availability section.

Ethics and Informed Consent

The data involving human participants were collected via an online experimental interface. Participation was entirely voluntary; individuals were presented with the study parameters and provided consent by actively choosing to initiate the experimental session. No personally identifiable information (PII) or IP addresses were collected, ensuring full participant anonymity. The study followed standard ethical principles for human subjects research, including voluntary participation, informed consent, and anonymized data handling. As this was a non-invasive behavioral study conducted by an independent researcher using anonymized data, formal Institutional Review Board (IRB) approval was not sought, but the protocol adhered to standard ethical principles for the protection of human subjects.

Conflict of Interest

The author declares no financial interest in any QRNG hardware, software platform, or related commercial product. This work was conducted independently and without external funding.

Data Availability Statement

The raw datasets, the frozen data manifest (SHA-256), and the Python analysis notebooks used to generate the findings in this study are available in the public repository at 10.5281/zenodo.18662941. The experimental source code (JavaScript), including the pre-specified hurstApprox and delta logic, is also provided on GitHub to ensure full transparency and replicability.

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